Requirements Gathering

Digital Egypt Pioneers Initiative (DEPI) | Data Engineering Final Project

Healthcare Data Pipeline Project

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1. Project Scope

The system will collect health data from separate CSV-based source systems, load it into a small analytical warehouse, apply transparent quality rules, and provide a Power BI report. The system is for education and portfolio demonstration. It is not a clinical decision system and must not be used to diagnose or treat patients.

2. Stakeholder Analysis

Stakeholder	Needs	How the project supports them
Technical instructor	A complete project with clear engineering decisions and evidence.	Documentation, demo, test results, and clear explanation of limitations.
Healthcare manager persona	Simple answers about health patterns and supplement activity.	Power BI overview, trends, and filterable dimensions.
Data analyst persona	Reliable tables and definitions for analysis.	Gold star schema, data dictionary, and documented measures.
Data engineer persona	Repeatable loads and visible data-quality issues.	Python ETL, SQL layers, audit table, reject table, and run instructions.
Project team	A manageable solution that can be completed and explained.	Small local stack, shared responsibilities, and staged milestones.

3. User Stories

1. As a healthcare manager, I want to see the number of users and health records so that I can understand the size of the data.

2. As a healthcare manager, I want to compare average sleep, glucose, and heart rate by date and age group so that I can see simple patterns.

3. As an analyst, I want to filter results by experiment and supplement so that I can compare groups.

4. As a data engineer, I want invalid and incomplete rows to be recorded with reasons so that data problems are visible.

5. As a project reviewer, I want the pipeline to run again with the same inputs so that I can trust the result.

6. As a new team member, I want setup and run instructions so that I can understand and reproduce the project.

4. Use Cases

Use case	Actor	Main steps	Expected result
Run ingestion	Data engineer	Select a batch, read the four CSV files, and write Bronze rows.	Raw rows are stored with source and batch metadata.
Clean and validate	Data engineer	Standardize types and values, apply rules, and separate rejects.	Silver tables and a reject report are produced.
Build Gold model	Data engineer	Create dimensions, facts, keys, and relationships.	Power BI-ready tables are available.
Review dashboard	Manager or analyst	Open the report, apply filters, and inspect measures.	The user can answer agreed business questions.
Check a load	Reviewer	Compare source, Silver, Gold, and reject counts.	The load is accepted or a clear issue is recorded.

5. Functional Requirements

ID	Requirement	Priority
FR-01	The system shall support the selected healthcare data sources after the team makes the final data-source decision.	Must
FR-02	The system shall support data from at least two source systems or clearly separated source areas.	Must
FR-03	The system shall preserve the original source values in a Bronze layer.	Must
FR-04	The system shall record the source name, file or table name, batch ID, load timestamp, and row count.	Must
FR-05	The system shall standardize column names, dates, data types, units, and other selected source values.	Must
FR-06	The system shall detect duplicates, missing values, invalid ranges, inconsistent formats, and broken relationships between source records.	Must
FR-07	The system shall store rejected records with a clear rejection reason and source information.	Must
FR-08	The system shall create cleaned Silver tables and business-ready Gold dimensions and fact tables.	Must
FR-09	The Gold layer shall use a suitable analytical model, such as a star schema, based on the final selected dataset.	Must
FR-10	The system shall make the Gold layer available for connection to Power BI.	Must
FR-11	The dashboard shall present useful healthcare trends, selected project KPIs, and data-quality results based on the final dataset.	Must
FR-12	The team shall document the selected data sources, transformations, data model, dashboard measures, and project limitations.	Must
FR-13	The team shall document possible future migration options, such as Azure SQL, Microsoft Fabric, or Databricks.	Should

6. Non-Functional Requirements

Area	Requirement
Usability	A beginner team member should be able to follow the README and run the pipeline without hidden manual steps.
Reliability	The pipeline should fail with a useful message and should not silently remove invalid rows.
Repeatability	The same input batch should produce the same row counts and measures.
Performance	The first version should finish within a few minutes on a normal student laptop.
Security	Only synthetic data will be used. No real patient identifiers, credentials, or private connection strings may be committed.
Maintainability	Cleaning rules, SQL scripts, and dashboard measures should have clear names and short comments.
Traceability	Every Silver and Gold load should be traceable to a source file and batch ID.
Accessibility	The dashboard should use readable titles, consistent colors, and avoid relying on color alone.

7. Assumptions and Open Questions

• Assumption: The existing health/supplement CSVs will be the first approved source.

• Assumption: SQL Server Developer or Express can be installed and used by at least one team member.

• Question: Will the instructor require a cloud platform, or is a local SQL Server solution accepted?

• Question: Should the final report focus on health behavior, supplement experiments, or hospital operations?

• Question: Which dashboard KPIs will be graded as mandatory?

• Question: Does the team need a real Gantt chart image, or is a Word table acceptable?

8. Acceptance Checklist

- All four sources are loaded successfully.
- Raw, clean, rejected, and Gold counts are documented.
- The Gold model has clear grain and one-to-many relationships.
- Dashboard measures match SQL validation queries.
- The report includes a data-quality page.
- The team can explain what is synthetic, what is transformed, and what is still a limitation.

References Used

- MITRE Synthea official site: https://synthetichealth.github.io/synthea/
- MITRE Synthea downloads and formats: https://synthetichealth.github.io/downloads.html
- Microsoft Fabric overview: https://learn.microsoft.com/en-us/fabric/fundamentals/microsoft-fabric-overview
- Fabric Data Warehouse: https://learn.microsoft.com/en-us/fabric/data-warehouse/data-warehousing
- Azure SQL Database documentation: https://learn.microsoft.com/en-us/azure/azure-sql/database
- Power BI star schema guidance: https://learn.microsoft.com/en-us/power-bi/guidance/star-schema
- Power BI overview: https://learn.microsoft.com/en-us/power-bi/fundamentals/power-bi-overview

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